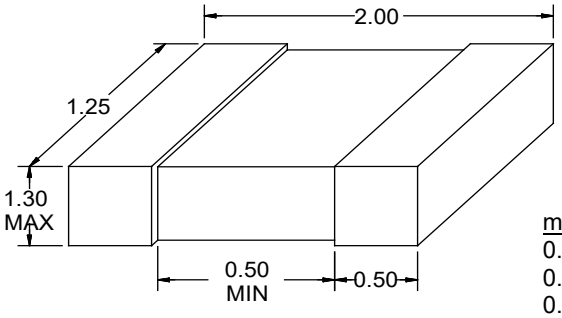
METRIC
MIL-PRF-55681/7D
24 March 2025
SUPERSEDING
MIL-PRF-55681/7C
10 October 2006

PERFORMANCE SPECIFICATION SHEET

CAPACITOR, CHIP, MULTIPLE LAYER, FIXED, CERAMIC DIELECTRIC, ESTABLISHED RELIABILITY AND NON-ESTABLISHED RELIABILITY, STYLE CDR31, METRIC

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-55681.



mm	Inches
0.20	.008
0.30	.012
0.50	.020
0.60	.023
1.25	.049
1.30	.051
2.00	.078

NOTES:
1. Dimensions are in millimeters.
2. Unless otherwise specified, tolerance is ± 0.20 mm (.008 inch).
3. Dimensions and tolerances are for bare chips. For solder coated terminations add 0.60 mm (.023 inch) to the positive length tolerance and 0.30 mm (.012 inch) to the positive width and thickness tolerance.
4. Inch equivalents are given for general information only.

FIGURE 1. Style CDR31 capacitors.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:54Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Capacitance value: See table I.

Capacitance tolerance: See table I.

Operating temperature range: -55° C to +125° C.

Product level designator: Non-established Reliability - C; Established Reliability - M, P, R, and S.

Marking: In accordance with MIL-PRF-55681.

TABLE I. Style CDR31 capacitor characteristics.

Part or Identifying Number (PIN) 1/	Capacitance pF	Capacitance tolerance	Rated temperature and voltage-temperature limits	Rated voltage (V_{dc})
CDR31BP1R0B - - -	1.0	B, C	BP	100
CDR31BP1R1B - - -	1.1	B, C	BP	100
CDR31BP1R2B - - -	1.2	B, C	BP	100
CDR31BP1R3B - - -	1.3	B, C	BP	100
CDR31BP1R5B - - -	1.5	B, C	BP	100
CDR31BP1R6B - - -	1.6	B, C	BP	100
CDR31BP1R8B - - -	1.8	B, C	BP	100
CDR31BP2R0B - - -	2.0	B, C	BP	100
CDR31BP2R2B - - -	2.2	B, C	BP	100
CDR31BP2R4B - - -	2.4	B, C	BP	100
CDR31BP2R7B - - -	2.7	B, C, D	BP	100
CDR31BP3R0B - - -	3.0	B, C, D	BP	100
CDR31BP3R3B - - -	3.3	B, C, D	BP	100
CDR31BP3R6B - - -	3.6	B, C, D	BP	100
CDR31BP3R9B - - -	3.9	B, C, D	BP	100
CDR31BP4R3B - - -	4.3	B, C, D	BP	100
CDR31BP4R7B - - -	4.7	B, C, D	BP	100
CDR31BP5R1B - - -	5.1	B, C, D	BP	100
CDR31BP5R6B - - -	5.6	B, C, D	BP	100
CDR31BP6R2B - - -	6.2	B, C, D	BP	100
CDR31BP6R8B - - -	6.8	B, C, D	BP	100
CDR31BP7R5B - - -	7.5	B, C, D	BP	100
CDR31BP8R2B - - -	8.2	B, C, D	BP	100
CDR31BP9R1B - - -	9.1	B, C, D	BP	100
CDR31BP100B - - -	10	F, J, K	BP	100
CDR31BP110B - - -	11	F, J, K	BP	100
CDR31BP120B - - -	12	F, J, K	BP	100
CDR31BP130B - - -	13	F, J, K	BP	100
CDR31BP150B - - -	15	F, J, K	BP	100
CDR31BP160B - - -	16	F, J, K	BP	100
CDR31BP180B - - -	18	F, J, K	BP	100
CDR31BP200B - - -	20	F, J, K	BP	100
CDR31BP220B - - -	22	F, J, K	BP	100
CDR31BP240B - - -	24	F, J, K	BP	100
CDR31BP270B - - -	27	F, J, K	BP	100
CDR31BP300B - - -	30	F, J, K	BP	100
CDR31BP330B - - -	33	F, J, K	BP	100
CDR31BP360B - - -	36	F, J, K	BP	100
CDR31BP390B - - -	39	F, J, K	BP	100

See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:54Z
Check the source to verify that this is the current version before use.

TABLE I. Style CDR31 capacitor characteristics. - continued.

PIN 1/	Capacitance pF	Capacitance tolerance	Rated temperature and voltage-temperature limits	Rated voltage (V_{dc})
CDR31BP430B - - -	43	F, J, K	BP	100
CDR31BP470B - - -	47	F, J, K	BP	100
CDR31BP510B - - -	51	F, J, K	BP	100
CDR31BP560B - - -	56	F, J, K	BP	100
CDR31BP620B - - -	62	F, J, K	BP	100
CDR31BP680B - - -	68	F, J, K	BP	100
CDR31BP750B - - -	75	F, J, K	BP	100
CDR31BP820B - - -	82	F, J, K	BP	100
CDR31BP910B - - -	91	F, J, K	BP	100
CDR31BP101B - - -	100	F, J, K	BP	100
CDR31BP111B - - -	110	F, J, K	BP	100
CDR31BP121B - - -	120	F, J, K	BP	100
CDR31BP131B - - -	130	F, J, K	BP	100
CDR31BP151B - - -	150	F, J, K	BP	100
CDR31BP161B - - -	160	F, J, K	BP	100
CDR31BP181B - - -	180	F, J, K	BP	100
CDR31BP201B - - -	200	F, J, K	BP	100
CDR31BP221B - - -	220	F, J, K	BP	100
CDR31BP241B - - -	240	F, J, K	BP	100
CDR31BP271B - - -	270	F, J, K	BP	100
CDR31BP301B - - -	300	F, J, K	BP	100
CDR31BP331B - - -	330	F, J, K	BP	100
CDR31BP361B - - -	360	F, J, K	BP	100
CDR31BP391B - - -	390	F, J, K	BP	100
CDR31BP431B - - -	430	F, J, K	BP	100
CDR31BP471B - - -	470	F, J, K	BP	100
CDR31BP511A - - -	510	F, J, K	BP	50
CDR31BP561A - - -	560	F, J, K	BP	50
CDR31BP621A - - -	620	F, J, K	BP	50
CDR31BP681A - - -	680	F, J, K	BP	50
CDR31BX471B - - -	470	K, M	BX	100
CDR31BX561B - - -	560	K, M	BX	100
CDR31BX681B - - -	680	K, M	BX	100
CDR31BX821B - - -	820	K, M	BX	100
CDR31BX102B - - -	1,000	K, M	BX	100
CDR31BX122B - - -	1,200	K, M	BX	100
CDR31BX152B - - -	1,500	K, M	BX	100
CDR31BX182B - - -	1,800	K, M	BX	100
CDR31BX222B - - -	2,200	K, M	BX	100
CDR31BX272B - - -	2,700	K, M	BX	100
CDR31BX332B - - -	3,300	K, M	BX	100
CDR31BX392B - - -	3,900	K, M	BX	100
CDR31BX472B - - -	4,700	K, M	BX	100
CDR31BX562A - - -	5,600	K, M	BX	50
CDR31BX682A - - -	6,800	K, M	BX	50
CDR31BX822A - - -	8,200	K, M	BX	50
CDR31BX103A - - -	10,000	K, M	BX	50
CDR31BX123A - - -	12,000	K, M	BX	50
CDR31BX153A - - -	15,000	K, M	BX	50
CDR31BX183A - - -	18,000	K, M	BX	50

1/ The complete PIN will include additional symbols to indicate capacitance tolerance, termination and product level.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:54Z
Check the source to verify that this is the current version before use.

Changes from previous issue: The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Army - CR
Navy - EC
Air Force – 85
DLA - CC

Review activities:
Navy - AS, MC
Air Force - 19

Preparing activity:
DLA - CC

(Project 5910-2025-014)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:54Z
Check the source to verify that this is the current version before use.